Assignment 1

Saturday, September 20, 2025 9:55 PM

A. $|0101\rangle = \frac{1}{\sqrt{2^4}}\left(|0\rangle + e^{2\pi i(0.1)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.10)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.101)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.0101)}|1\rangle\right)$

$\text{Sum}(k) \rightarrow \frac{1}{\sqrt{2^4}}\left(|0\rangle + e^{2\pi i(0.1)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.10)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.1011)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.01011)}|1\rangle\right)$

$\text{IQFT} \rightarrow |a+k\rangle \rightarrow |1011\rangle$

B. $|\psi_b\rangle \downarrow$ $k=4$, $N=5$, $a=3$

$|00110\rangle$ $\text{QFT} \rightarrow \frac{1}{\sqrt{2^4}}\left(|0\rangle + e^{2\pi i(0.0)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.00)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.001)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.0011)}|1\rangle\right)|0\rangle$

$\text{Sum}(k) \rightarrow \frac{1}{\sqrt{2^4}}\left(|0\rangle + e^{2\pi i(0.0)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.01)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.011)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.0111)}|1\rangle\right)|0\rangle$

$\text{Sum}(-N) \rightarrow \frac{1}{\sqrt{2^4}}\left(|0\rangle + e^{2\pi i(0.0)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.00)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.001)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.0010)}|1\rangle\right)|0\rangle$

$\text{IQFT} \rightarrow |00100\rangle \rightarrow Cx|00100\rangle \rightarrow |00100\rangle$

$|\psi_c\rangle \downarrow$

$|00100\rangle$ $\text{QFT} \rightarrow \frac{1}{\sqrt{2^4}}\left(|0\rangle + e^{2\pi i(0.0)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.00)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.001)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.0010)}|1\rangle\right)|0\rangle$

$\text{Sum}(N) \rightarrow$ "not activated"

$\text{Sum}(-k) \rightarrow \frac{1}{\sqrt{2^4}}\left(|0\rangle + e^{2\pi i(1.1)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(1.11)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(1.110)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(1.0110)}|1\rangle\right)|0\rangle$

$\text{IQFT} \rightarrow |11100\rangle \rightarrow Cx\ \text{ctrl}=0\ |11100\rangle \rightarrow |11100\rangle$

$|\psi_d\rangle \downarrow$

$\text{QFT} \rightarrow \frac{1}{\sqrt{2^4}}\left(|0\rangle + e^{2\pi i(1.1)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(1.11)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(1.110)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(1.1100)}|1\rangle\right)|0\rangle$

$\text{Sum}\ k \rightarrow \frac{1}{\sqrt{2^4}}\left(|0\rangle + e^{2\pi i(0.0)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.00)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.001)}|1\rangle\right)\left(|0\rangle + e^{2\pi i(0.0010)}|1\rangle\right)|0\rangle$

$\text{IQFT} \rightarrow |00100\rangle$

C) $|011000\rangle$ $N=5$, $a=3$, $k=4$

$|\psi_b\rangle \downarrow$

$\text{Add}_{in}(2^1, N) \rightarrow |011\rangle|2+0 \bmod 5\rangle \rightarrow |011\rangle|010\rangle$

$\text{Add}_{in}(2^0, N) \rightarrow |011\rangle|2+1 \bmod 5\rangle \rightarrow |011\rangle|011\rangle$

$|\psi_c\rangle \downarrow$

$\text{Add}_{in}(k, N) \rightarrow |011\rangle|3+4 \bmod 5\rangle \rightarrow |011\rangle|010\rangle$

D) $|011010\rangle$ $N=5$, $a=3$, $b=2$, $k=4$

$\text{Add}_{in}(2^2, N) \rightarrow$ No activation

$\text{Add}_{in}(2^1, N) \rightarrow |011\rangle|2+2 \bmod 5\rangle \rightarrow |011\rangle|100\rangle$

$\text{Add}_{in}(2^0, N) \rightarrow |011\rangle|4+1 \bmod 5\rangle \rightarrow |011\rangle|000\rangle$

$|011000\rangle$

E) $|011010000\rangle$ $N=5$, $a=3$, $b=2$, $k=4$

$\text{Add}_{in(a)}(2^2, N) \rightarrow$ Not activated

$\text{Add}_{in(a)}(2^1, N) \rightarrow |011010\rangle|2+0 \bmod 5\rangle \rightarrow |011010\rangle|010\rangle$

$\text{Add}_{in(a)}(2^0, N) \rightarrow |011010\rangle|1+2 \bmod 5\rangle \rightarrow |011010\rangle|011\rangle$

$\text{Add}_{in(b)}(2^2, N) \rightarrow$ Not activated

$\text{Add}_{in(b)}(2^1, N) \rightarrow |011010\rangle|3+2 \bmod 5\rangle \rightarrow |011010\rangle|000\rangle$

$\text{Add}_{in(b)}(2^0, N) \rightarrow$ Not activated

$|011010000\rangle$

F) $|011011\rangle$ $a=3$, $b=3$, $k=2$, $N=5$

$\text{Add}_{in}(k2^2, N) \rightarrow$ Not activated

$\text{Add}_{in}(k2^1, N) \rightarrow |011\rangle|2\cdot 2+3 \bmod 5\rangle \rightarrow |011\rangle|010\rangle$

$\text{Add}_{in}(k2^0, N) \rightarrow |011\rangle|2\cdot 1+2 \bmod 5\rangle \rightarrow |011\rangle|100\rangle$

$|011100\rangle$

G) $|011000\rangle$ $a=3$, $k=2$, $N=5$

$\text{Mult}_{out}(k, N) \rightarrow |011\rangle|3\cdot 2+0 \bmod 5\rangle = |011\rangle|001\rangle$

$\text{Swap} \rightarrow |001\rangle|011\rangle$

$\text{Mult}_{out}(k^{-1}, N)^{+} \rightarrow |2\cdot 3 \bmod 5\rangle|000\rangle \rightarrow |001\rangle|000\rangle$

$|001000\rangle$

H) $|011001000\rangle$ $N=5$, $a=3$, $b=1$

$\text{Mult}_{out}(2^2, N) \rightarrow$ Not activated

$\text{Mult}_{out}(2^1, N) \rightarrow |011001\rangle|0+2\cdot 1 \bmod 5\rangle \rightarrow |011001\rangle|010\rangle$

$\text{Mult}_{out}(2^0, N) \rightarrow |011001\rangle|2+1\cdot 1 \bmod 5\rangle \rightarrow |011001\rangle|011\rangle$

$|011001011\rangle$

J) $|011\ 111000\rangle$ $N=5$, $x=3$, $a=2$

$\text{Mult}_{in}(a^{2^2}) \rightarrow$ Not activated

$\text{Mult}_{in}(a^{2^1}) \rightarrow |011\rangle|4\cdot 7 \bmod 5\rangle|000\rangle \rightarrow |011\rangle|011\rangle|000\rangle$

$\text{Mult}_{in}(a^{0}) \rightarrow |011\rangle|3\cdot 1 \bmod 5\rangle|000\rangle \rightarrow |011\rangle|011\rangle|000\rangle$

$|011011000\rangle$